



CYBER SECURITY

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TODAY'S TOPICS

1. Cyberspace

2. Dangers of the internet

3. How do websites protect you?

4. How do you protect yourself?



WHAT IS CYBERSPACE?

- Cyberspace is anything that has to do with the Internet.
- We use the Internet to communicate and stay connected to our friends and family.
- We play videogames, download music, and watch TV shows and movies
- Grownups do things like banking, shopping, and other important everyday activities online too.

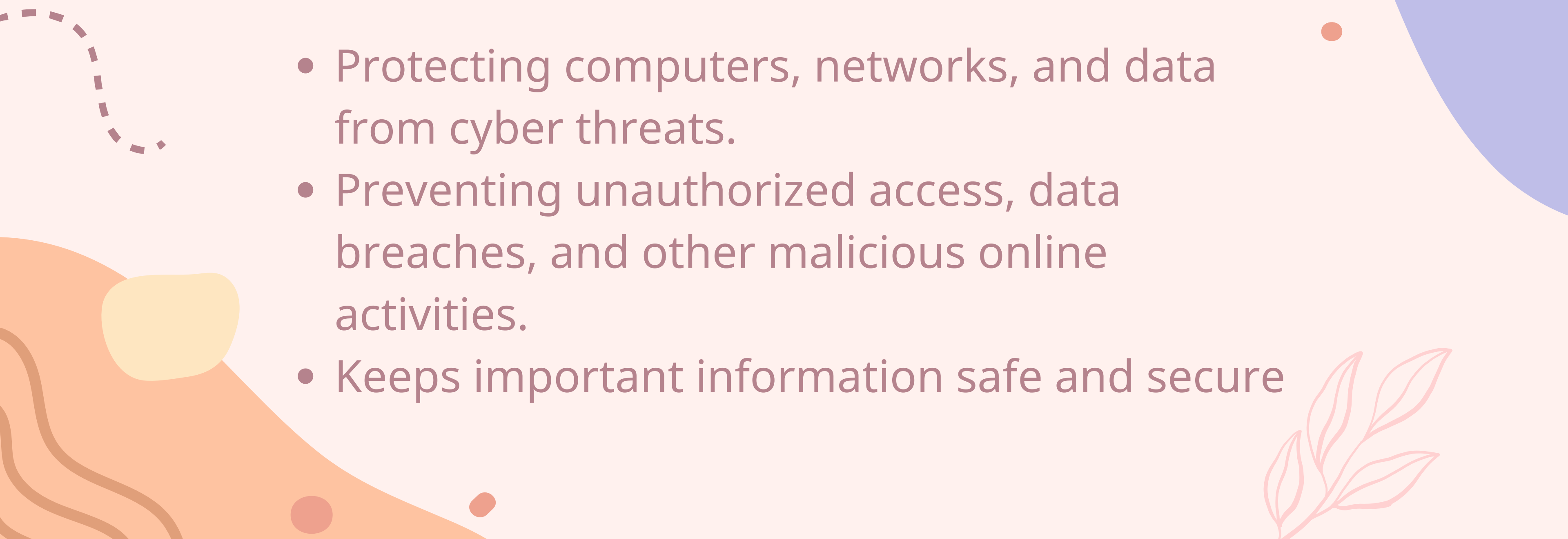


WHY DO WE NEED TO PROTECT CYBERSPACE?

- Protect private information from identity theft
- Financial security
- Preventing cyberbullying and online harassment
- Protecting personal devices



WHAT IS CYBERSECURITY?

- 
- Protecting computers, networks, and data from cyber threats.
 - Preventing unauthorized access, data breaches, and other malicious online activities.
 - Keeps important information safe and secure

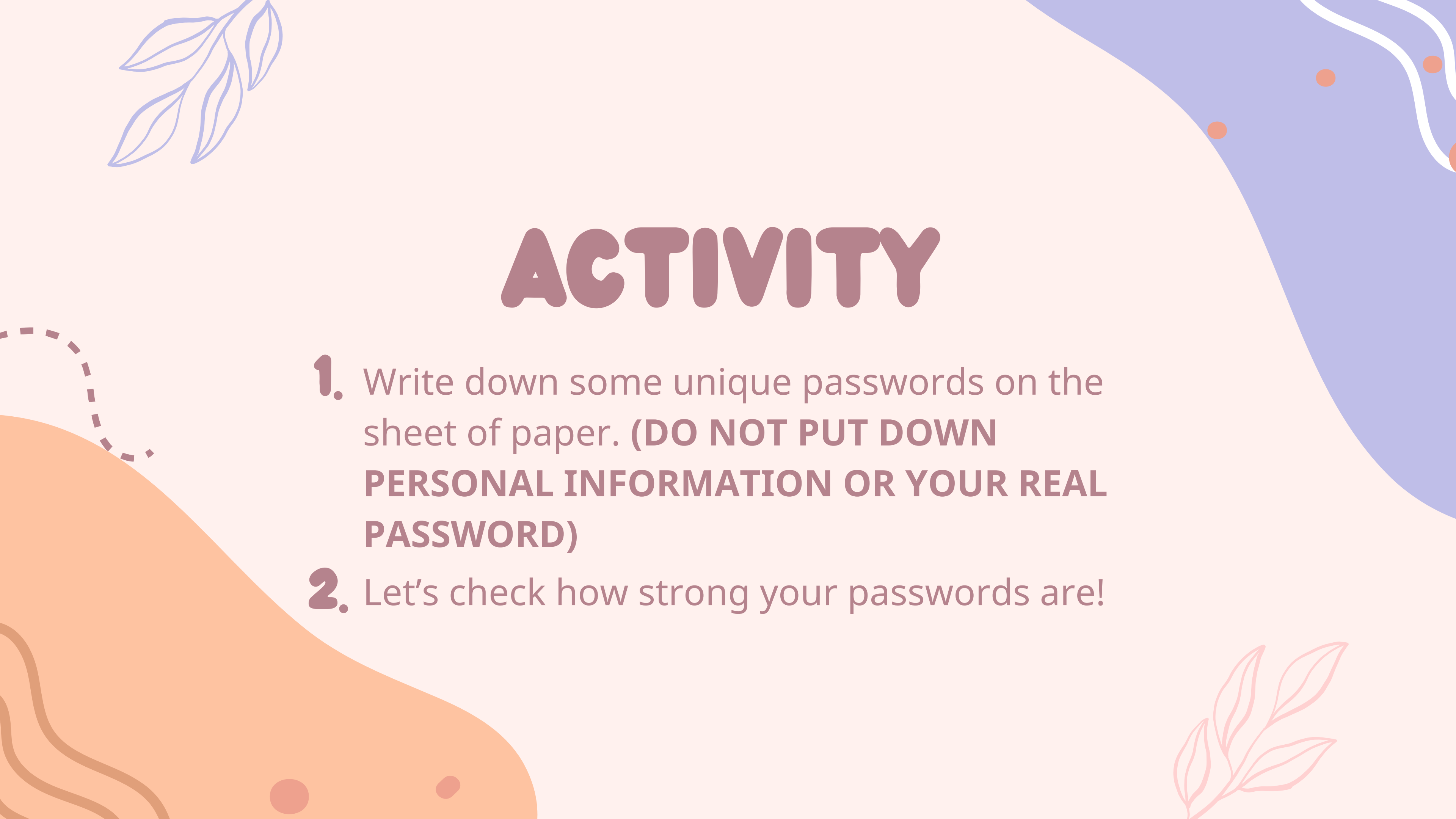
IMPORTANCE OF A GOOD PASSWORD





IMPORTANCE OF A GOOD PASSWORD

- Having a good password is important because it's like having a strong lock on your online accounts
- If you use a weak password, it makes it super easy for hackers and cybercriminals to steal your information.



ACTIVITY

- 1.** Write down some unique passwords on the sheet of paper. **(DO NOT PUT DOWN PERSONAL INFORMATION OR YOUR REAL PASSWORD)**
- 2.** Let's check how strong your passwords are!



USE STRONG, UNIQUE PASSWORDS

- Using a strong password makes it harder for hackers to guess your password at random
- Using unique passwords makes it so that if one of your passwords is found then a hacker won't be able to access other accounts of yours
- Change your password regularly because it will eventually get cracked

WHAT MAKES A STRONG PASSWORD?

- Has numbers letters and special characters
- Doesn't relate to any personal information
- At least 10 characters long (the longer the better)
- This is important because stolen passwords make up for 80% of data breaches

Weak Password	OK Password	Strong Password
rose	rosemary1033	Rosemary1033@-@
isum	isumsoft\$\$	iSumsoft100\$\$*_*
jellyfish	jelly22fish	Jelly22fi\$h

DANGERS OF THE INTERNET



WHAT IS SOCIAL ENGINEERING?

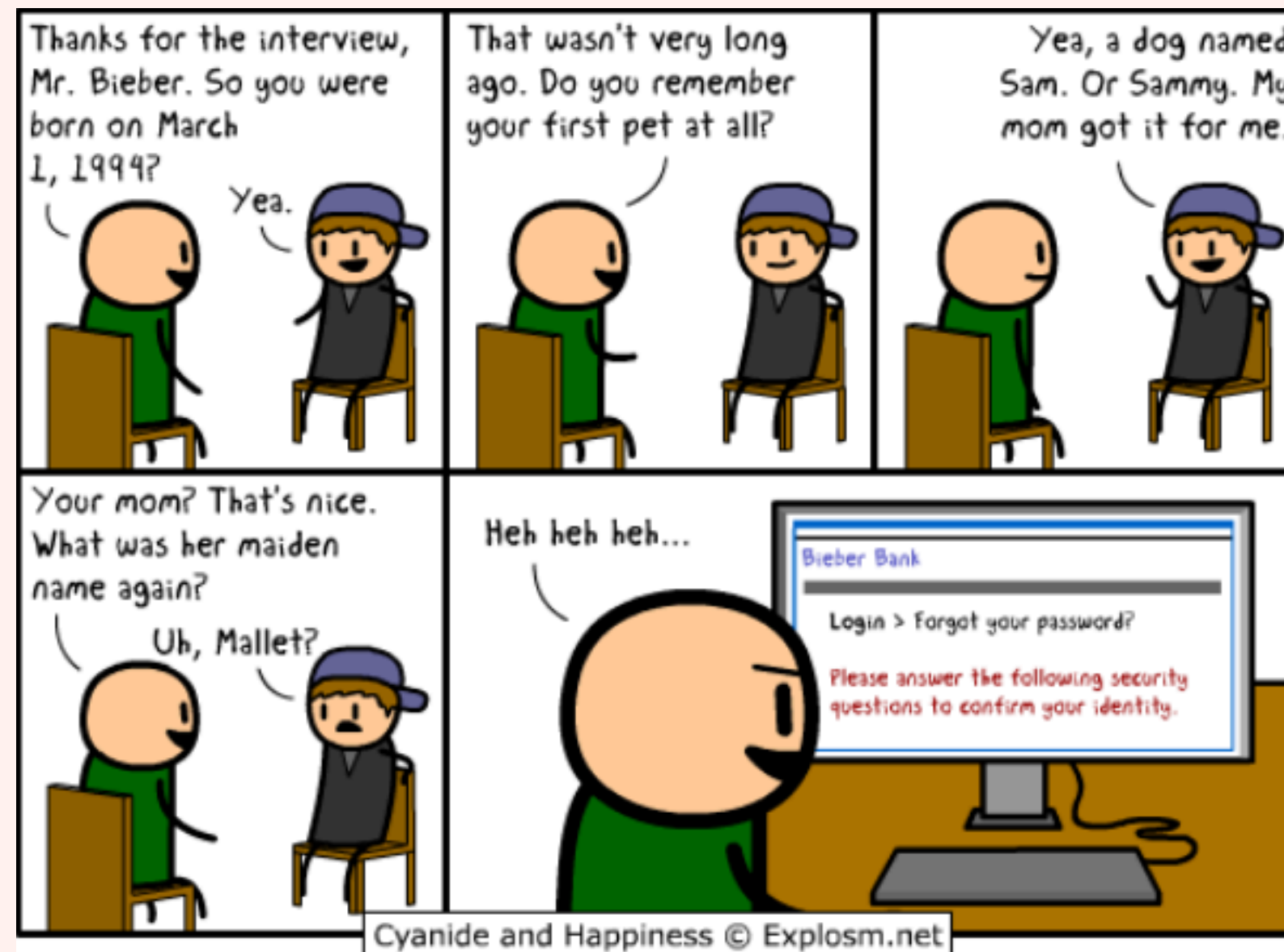


SOCIAL ENGINEERING

- A technique used to manipulate people into giving up information or performing action that compromise their security

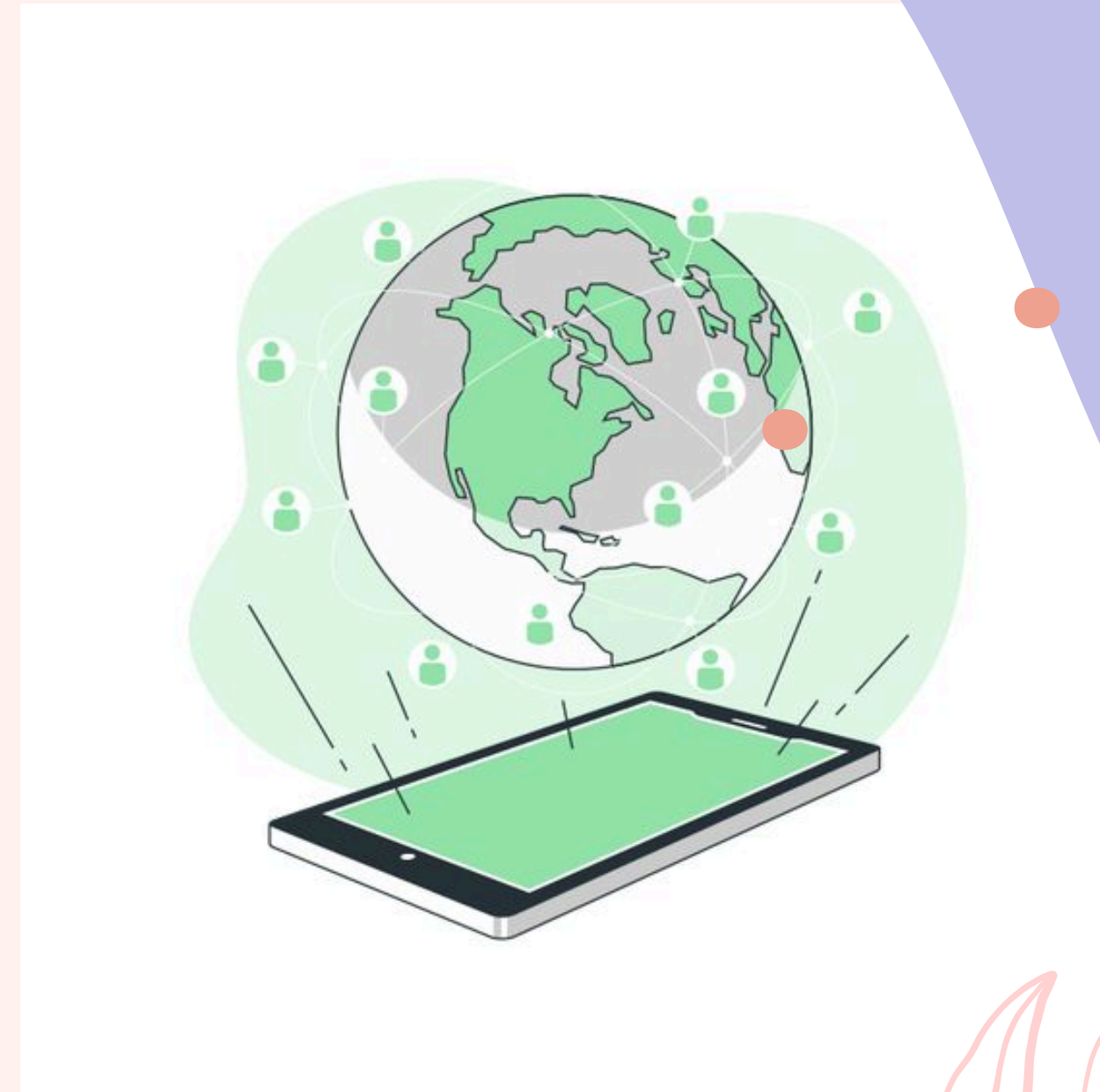


KIND OF LIKE....



What seems to be happening to the bald man?

**SHARE WITH
CARE**



BEFORE YOU SHARE, REMEMBER...

- Your online actions can have real world consequences*
 - The pictures you post or send and the words you write can affect the people in your life.



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BEFORE YOU SHARE, REMEMBER...

- Once you post something, you can't take it back.
 - Even for sending messages. If you delete it, copies might exist on other people's devices.
 - The post could live on the internet permanently!



BEFORE YOU SHARE, REMEMBER...

- What you post could have a bigger “audience” than you think.
 - It is impossible to have complete control over who sees your pictures, videos, social media profiles, comments, or texts.
 - Think before you send. How would you feel if your parents, teachers, or coaches find it?



DATA BREACHES

What is a data breach?

- Someone steals your information or data
- This is done through hacking or other cyber attacks

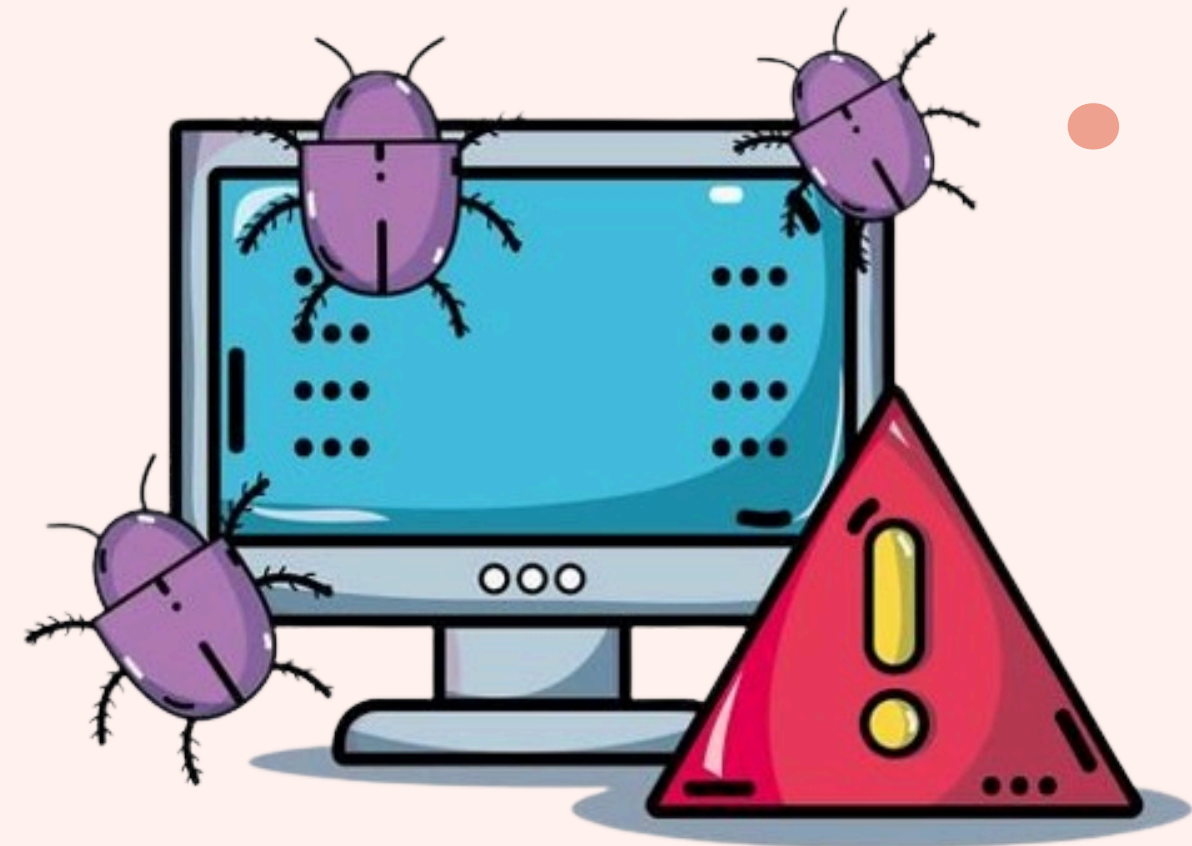
What are the dangers?

- Financial losses
- Unauthorized access of personal information



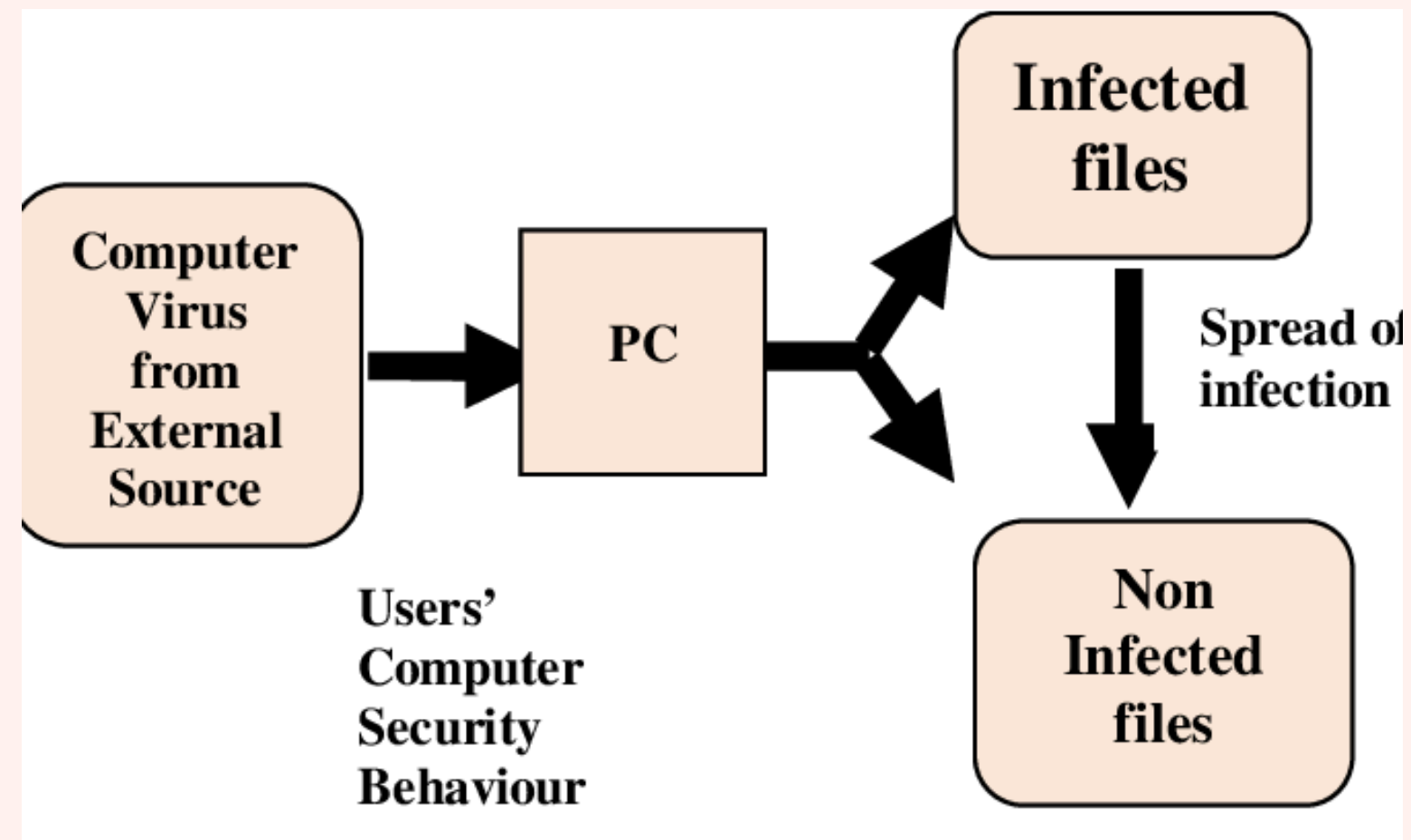
TYPES OF MALWARE

- Virus
- Worms
- Ransomware
- You don't want these!



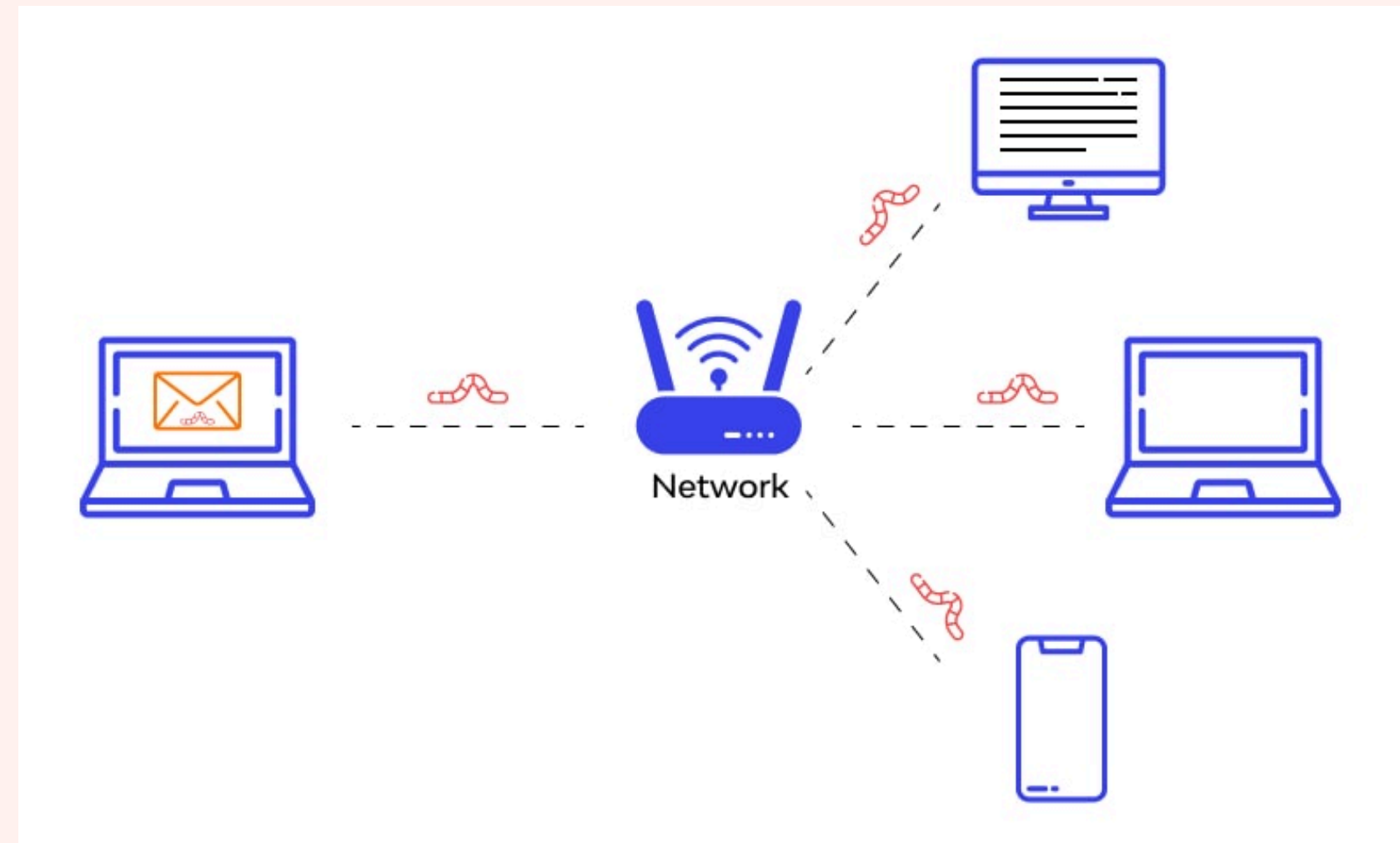
VIRUS

- Attaches to a file and spreads when the infected file is executed.



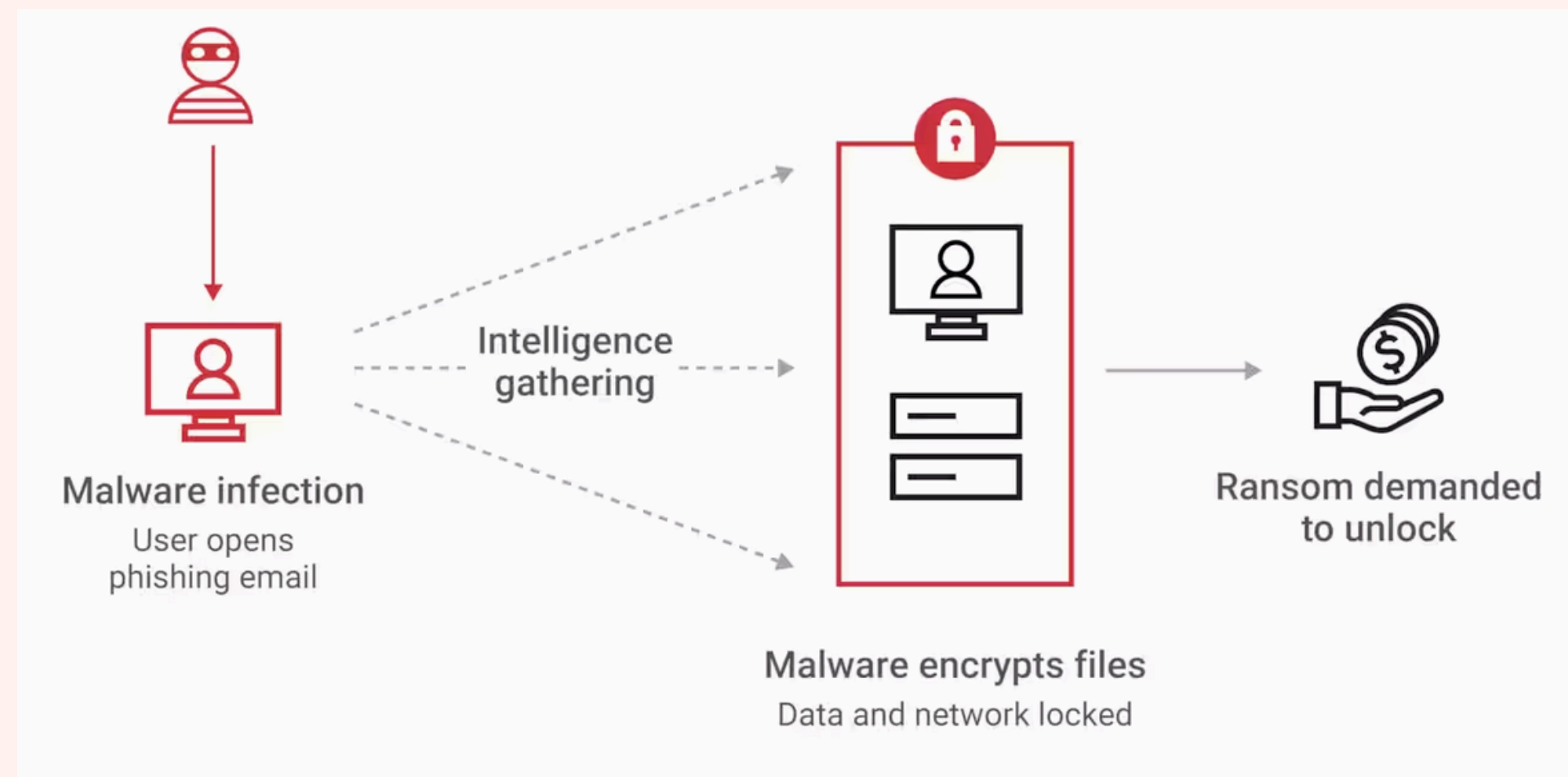
WORM

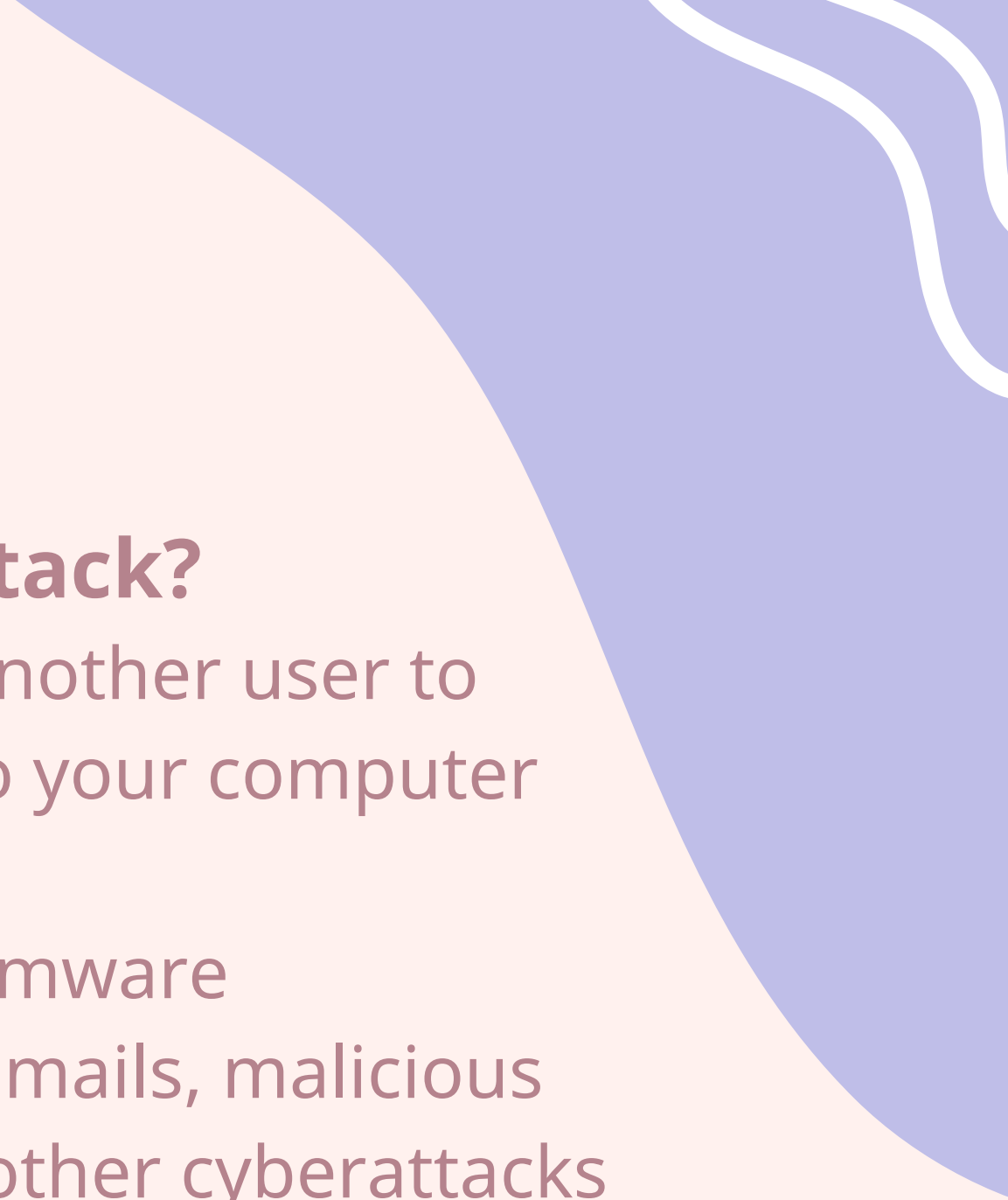
- Spread from one device to another through email systems or networks.



RANSOMWARE

- Scrambles your files and demands payment in exchange to descramble them.





MALWARE ATTACKS

What is a Malware Attack?

- Malware can allow a another user to harm or gain access to your computer through software
- Viruses, worms, ransomware
- This is done through emails, malicious websites, hacking, or other cyberattacks

What are the dangers?

- Loss of data
 - Financial losses
 - Destruction of your computers software
- 

ACTIVITY: JAKE AND THE BAD VIRUS

Jake was using the Internet on his family's home computer to research a school project on dolphins. He just finished finding the perfect article to add to his research and was about to log off the computer. Before shutting down Jake messages including one from from NO1GRANDPA@HAPPYMAIL.COM, which he recognizes as his Grandfather's email address, and another from SWEEPSTAKES@WHOKNOWS.COM with a subject line that reads --YOU'VE WON \$5,000.

ACTIVITY: JAKE AND THE BAD VIRUS

Jake skips his Grandfather's email and quickly opens the sweepstakes email to collect his prize. After doing so, Jake receives a message instructing him to provide personal information to verify his identity. Jake provides his full name, birthday, home address and phone number. He hits submit and instantly starts receiving messages saying his computer has been infected by a virus.

ACTIVITY: JAKE AND THE BAD VIRUS

Jake panics because he doesn't want to get into trouble. He quickly logs off, shuts off his computer, and goes to bed, hoping his parents won't know what happened in the morning.

- What were Jake's mistakes?
- What should Jake have done?

PHISHING



- Phishing is a way that hackers can take advantage of you by deceiving you.
- Hackers may do this by sending out emails claiming to be a family member of yours or a well known company.
- They may then ask for personal information which they can then use for their own good.
- Common things they ask for are your social security number, credit card number, or passwords.

The background is a light pink color with various decorative elements. In the top left, there is a blue line-art illustration of a leafy branch. In the top right, there is a large blue abstract shape with a yellow circle inside and several orange dots. In the bottom left, there is an orange abstract shape with a yellow circle inside and several orange dots. In the bottom right, there is a pink line-art illustration of a leafy branch. Two dashed lines, one in the top center and one in the bottom center, form a wavy, horizontal shape.

ACTIVITY

**PHISHING OR NOT:
[HTTPS://WWW.PHISHINGBOX.COM/PHISHING-TEST/](https://www.phishingbox.com/phishing-test/)**

SCAMS



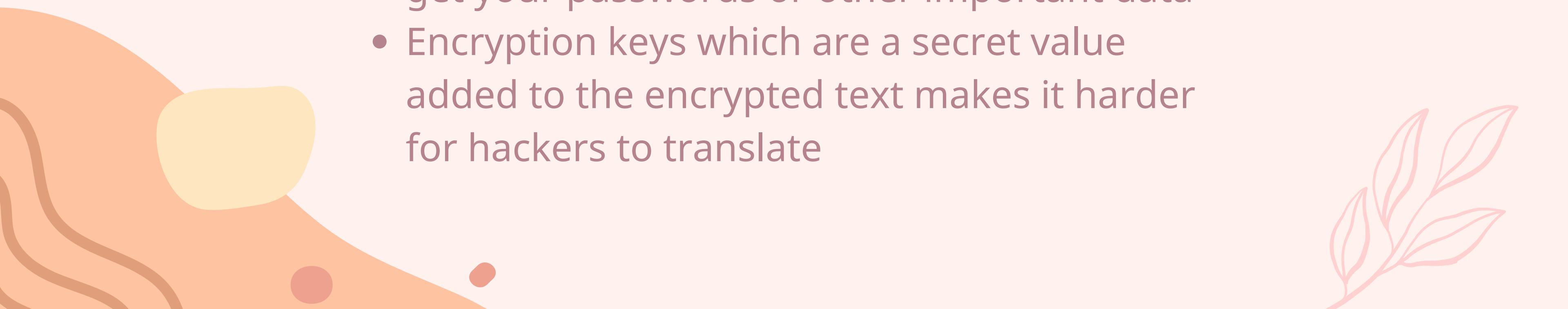
- Scams are another way that people can get ahold of your information by deceiving you.
- Scams can be things such as investment scams or prize scams.
- They can do this to get ahold of your personal information and make money.
- Common things they ask for is once again your social security number, credit card number, or passwords.



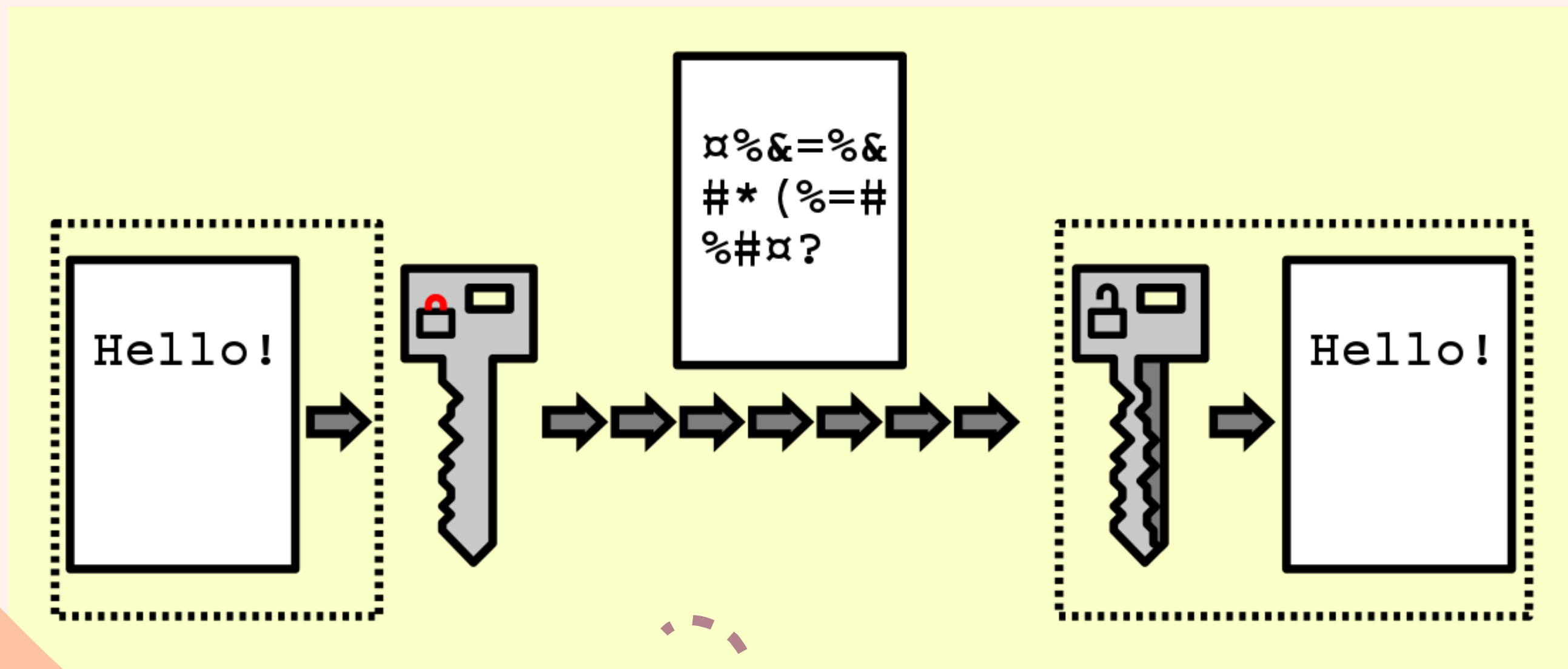
HOW DO WEBSITES PROTECT YOU?



ENCRYPTION

- Encryption changes your plain text into a scrambled unreadable form
 - By doing this it makes it harder for hackers to get your passwords or other important data
 - Encryption keys which are a secret value added to the encrypted text makes it harder for hackers to translate
- 

ENCRYPTION DIAGRAM

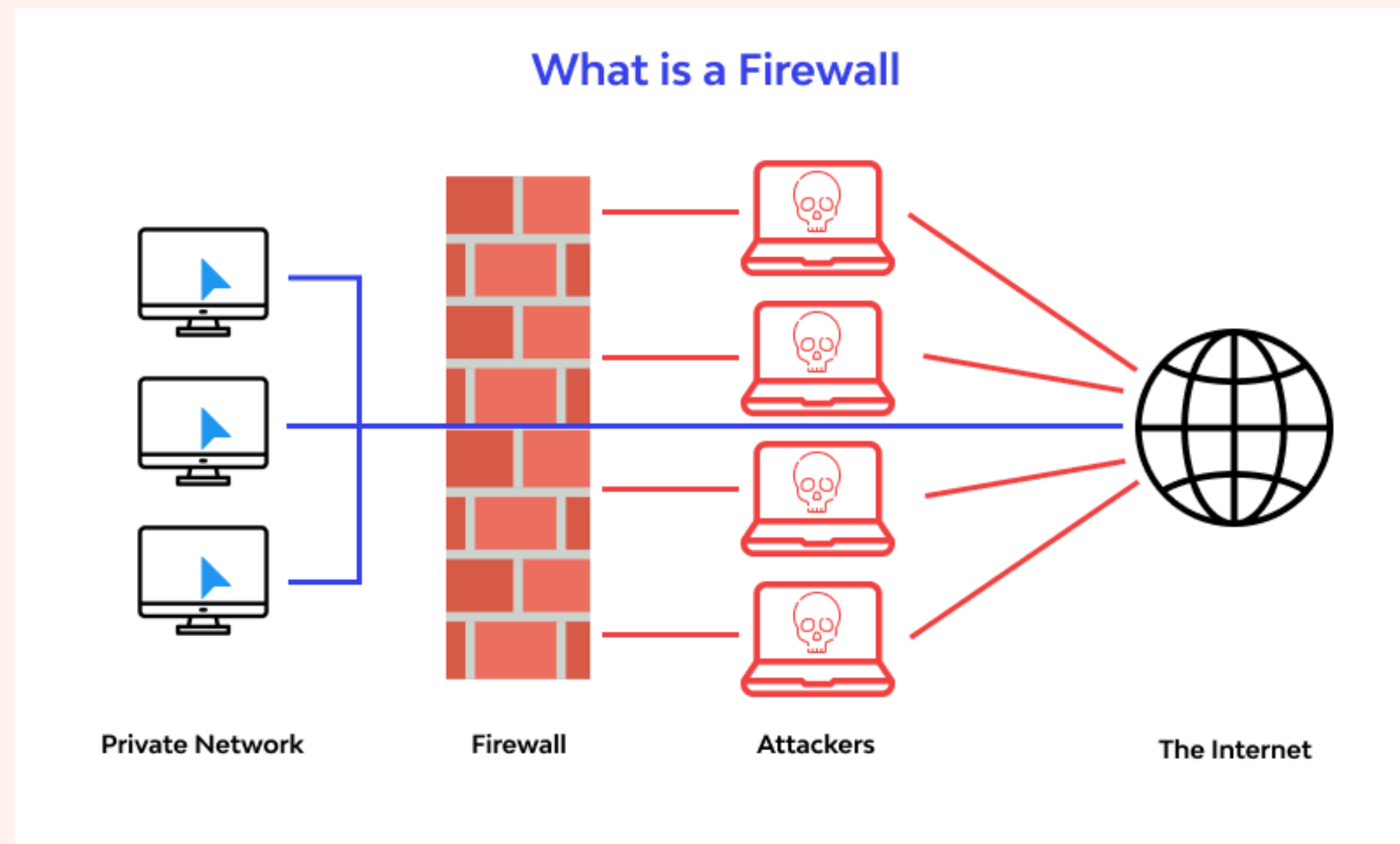




FIREWALLS

- A fire wall is like a security guard for your house.
- They will examine anyone who wants to enter or leave.
- Firewalls do the same by examining the incoming data packets to your network.
- This prevents hackers from performing an attack.

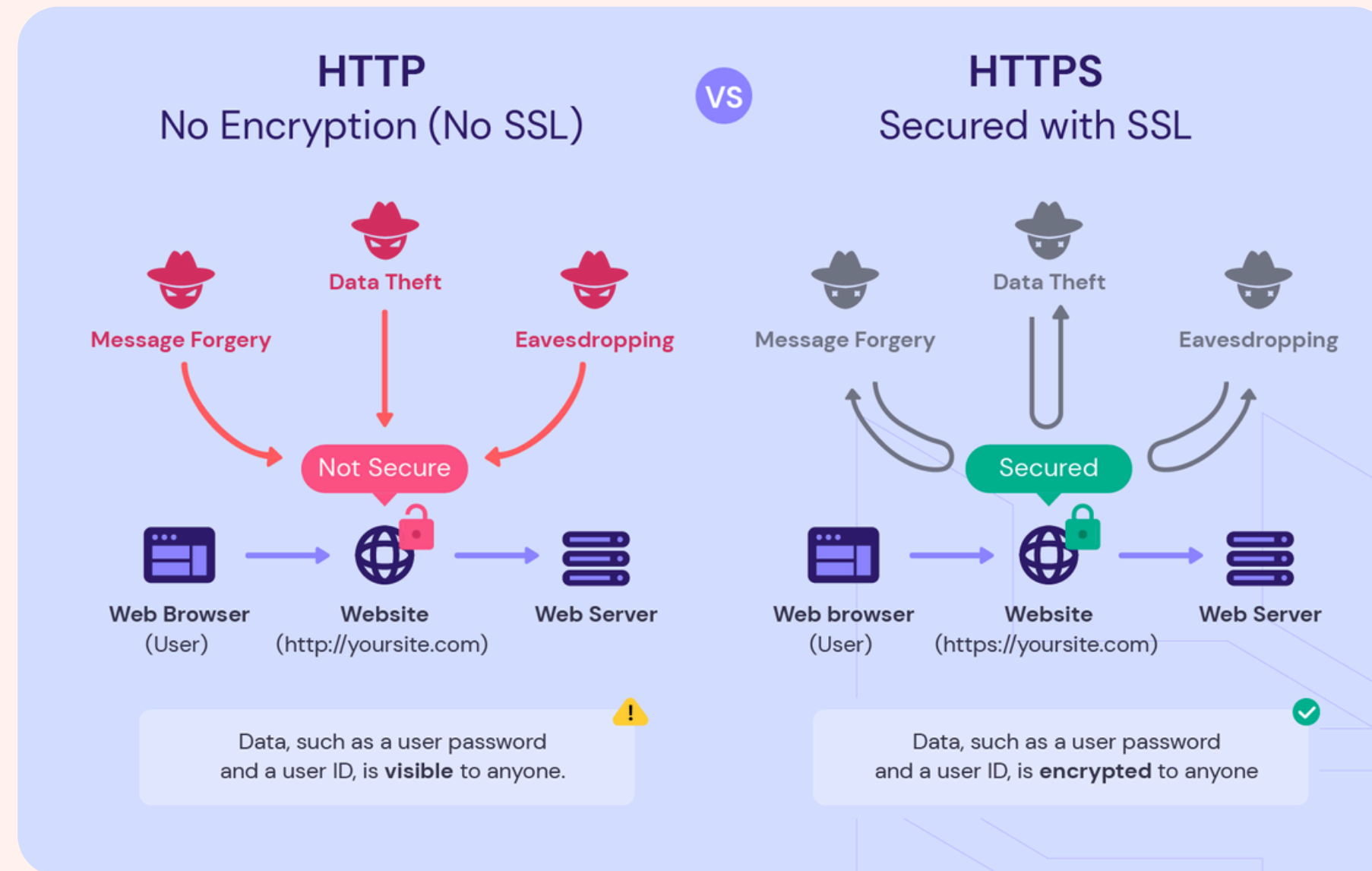
FIREWALL DIAGRAM



WHAT ARE SECURE SITES?



HTTP VS HTTPS



HTTP and HTTPS are like two different ways of sending mail.

HTTP

- Hypertext Transfer Protocol
- HTTP is like sending a postcard through the mail. It's quick and easy because there's no envelope, but anyone who handles it along the way can read what's written on it



HTTPS

- Hypertext Transfer Protocol Secure
- HTTPS is like sending a letter in a sealed envelope. Even though it may take more effort to seal it, it ensures that only the intended recipient can open and read the contents, keeping your message private and secure.
- You can check if your on a website with https by looking for https:// in the website url

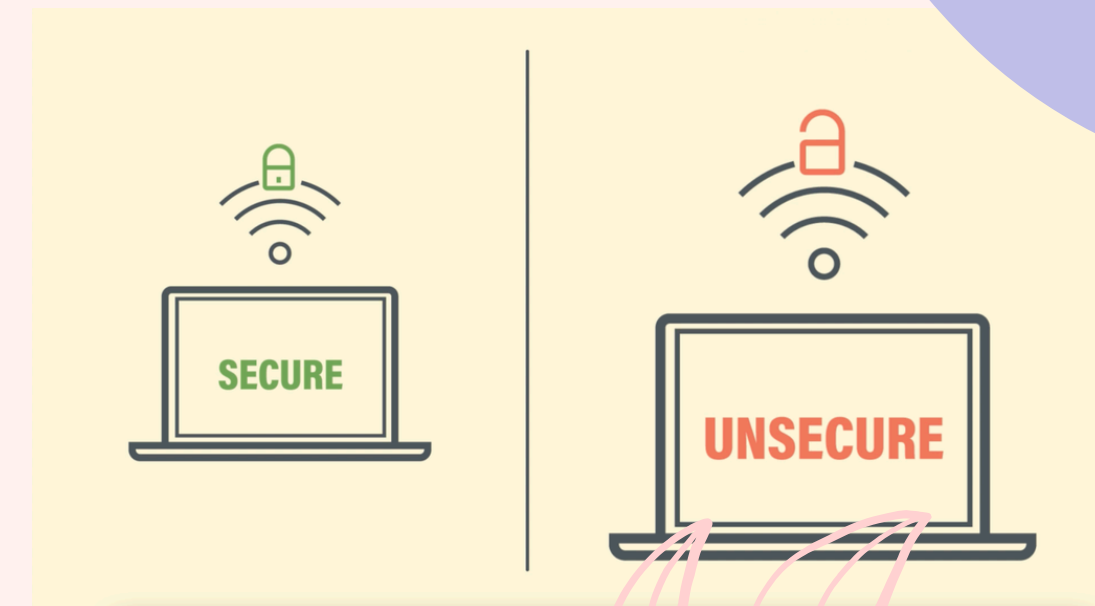


The background is a light pink color. It features several decorative elements: a blue floral motif in the top left, a blue wavy shape with orange dots in the top right, an orange wavy shape with a yellow circle in the bottom left, and a pink floral motif in the bottom right. There are also three dashed lines in a wavy pattern, one in the top center, one on the left side, and one in the bottom center.

**HOW CAN YOU
PROTECT
YOURSELF?**

USE SECURE CONNECTIONS

- Always use secure Wi-Fi networks and HTTPS enabled websites
- You can check if your Wi-Fi network is secured by encryption if there is a padlock symbol. If your Wi-Fi network is unsecure you should stop using the Wi-Fi to ensure your safety





INSTALL ANTIVIRUS

- If your device doesn't have it already, installing antivirus software can help protect you from malware.
- You can do this by looking up a youtube video and following the instructions.
- The antivirus software monitors your computer and searches for potential malware and it will help you get rid of it.



ENABLE TWO-FACTOR AUTHENTICATION (2FA)

- 2FA makes it so you have to pass two authentication tests in order to log into a account
 - This makes it so the hacker would have to have access to two separate passwords which is very unlikely
 - You can enable 2FA on many websites in the settings if you follow the instructions
- 



BE CAUTIOUS OF SUSPICIOUS EMAILS AND LINKS

- Many emails and links can lead to your information being compromised
 - These emails can be impersonating real organizations which can lead to data breaches
 - Clicking on links can cause your device to download malware
 - So it's best to not click on links or open emails from a unknown user
- 



ACTIVITY: SPOT THE FAKE ADDRESSES

help@spotify-support.net help@sp0tify.com

service@paypa1.com service@paypal-support.info

• support@apple.support help@amaz0n.com

• help@amaz0n-support.com





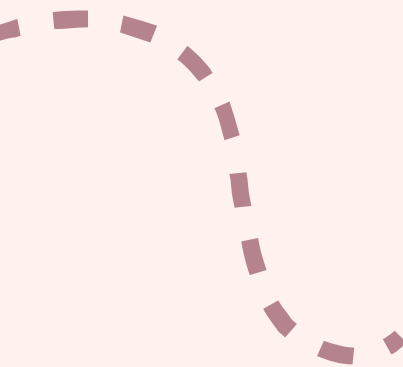
ALL ARE FAKE

help@spotify-support.net help@sp0tify.com
service@paypa1.com service@paypal-support.info
support@apple.support help@amaz0n.com
help@amaz0n-support.com



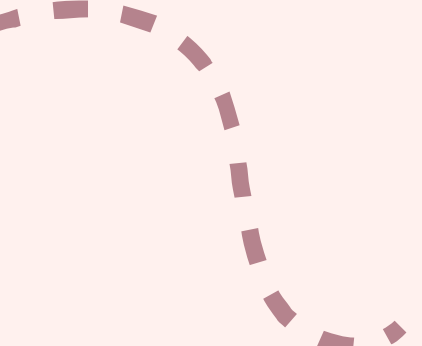


HOW TO AVOID SCAMS AND PHISHING

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- If something seems too good to be true it probably is.
 - Following the previous steps: Enable 2FA and be cautious of suspicious emails and links.
 - Check the source of information and make sure they are who they say they are.
- 



AVOIDING VIRUSES WORMS AND RANSOMWARE

- 
- 
- Keep your software up-to-date
 - Install a anti-virus software
 - Only download from sources you can trust
 - Be wary of emails from people you don't know
 - Avoid clicking on links from websites or people you can't trust
- 



ACTIVITY

Go to this website:

<https://sos.fbi.gov/en/>



ACTIVITY

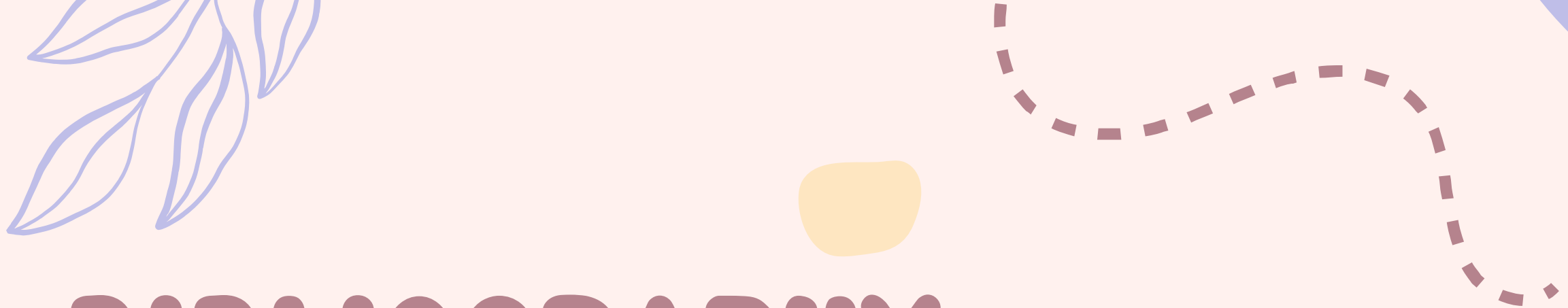
Ai or human?

<https://humanornot.so>



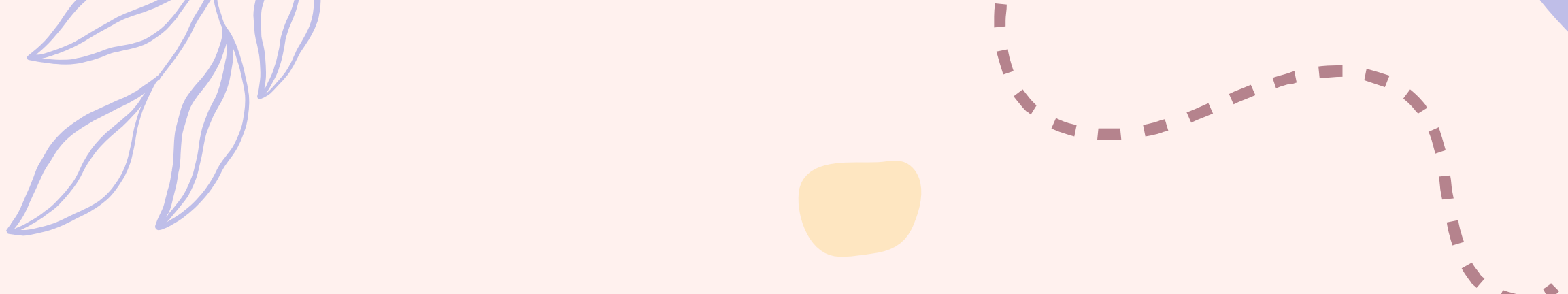
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 - <https://www.pinterest.com>
 - And any other site we may have accidentally stolen pictures from :)
- 



QUESTIONS?
THANK YOU!